

Exercise 1

On the basis of the Consumer Survey conducted by the Bureau of Labor Statistics, US Department of Labour your lecturer estimated a tobit model for sweets expenditures (*sweets*). The list of regressors consisted of:

- *fam_size* – family size
- *sex_ref* - sex of the reference person in the household
- *age_ref* - age of the reference person in the household (1 for men, 2 for women)
- *vehq* - number of vehicles in the household

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xi: tobit sweets fam_size i.sex_ref age_ref vehq, ll
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Tobit regression	Number of obs	=	2746
	LR chi2(4)	=	96.30
	Prob > chi2	=	0.0000
Log likelihood = -5379.949	Pseudo R2	=	0.0089

sweets	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
fam_size	1.553851	.1883082	8.25	0.000	1.184611	1.923092
_Isex_ref_2	1.222848	.4794174	2.55	0.011	.2827921	2.162904
age_ref	.0413944	.014952	2.77	0.006	.0120762	.0707126
vehq	.2878752	.2356275	1.22	0.222	-.1741502	.7499005
_cons	-9.815816	1.101803	-8.91	0.000	-11.97626	-7.655369
/sigma	10.79219	.245705			10.3104	11.27397

Obs. summary: 1569 left-censored observations at sweets<=0
 1177 uncensored observations
 0 right-censored observations

Source	SS	df	MS	Number of obs	=	2746
Model	1990.77508	4	497.693769	F(4, 2741)	=	14.41
Residual	94696.736	2741	34.5482437	Prob > F	=	0.0000
Total	96687.511	2745	35.223137	R-squared	=	0.0206
				Adj R-squared	=	0.0192
				Root MSE	=	5.8778

sweets	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
fam_size	.5656429	.0894017	6.33	0.000	.3903414	.7409445
_Isex_ref_2	.4387352	.2266614	1.94	0.053	-.0057092	.8831796
age_ref	.0122419	.006882	1.78	0.075	-.0012525	.0257364
vehq	.1212172	.1119189	1.08	0.279	-.0982367	.3406711
_cons	.475403	.4856015	0.98	0.328	-.4767789	1.427585

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. dtobit
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Marginal Effects: Latent Variable

variable	dF/dx	Std. Err.	z	P> z	X_at	[	95% C.I.	]
fam_size	1.553851	.1883082	8.25	0.000	2.50255	1.18477	1.92293	
_Isex_ref_2*	1.222848	.4794174	2.55	0.011	0 --> 1	.283207	2.16249	
age_ref	.0413944	.014952	2.77	0.006	50.1176	.012089	.0707	
vehq	.2878752	.2356275	1.22	0.222	1.90058	-.173946	.749697	
_cons	-9.815816	1.101803	-8.91	0.000	1	-11.9753	-7.65632	

Marginal Effects: Unconditional Expected Value

variable	dF/dx	Std. Err.	z	P> z	X_at	[	95% C.I.	]
fam_size	.6233747	.0755456	8.25	0.000	2.50255	.475308	.771441	
_Isex_ref_2*	.4909579	.1923329	2.55	0.011	0 --> 1	.113992	.867923	
age_ref	.0166066	.0059984	2.77	0.006	50.1176	.00485	.028363	
vehq	.1154899	.0945291	1.22	0.222	1.90058	-.069784	.300764	
_cons	-3.937913	.4420216	-8.91	0.000	1	-4.80426	-3.07157	

Marginal Effects: Conditional on being Uncensored

variable	dF/dx	Std. Err.	z	P> z	X_at	[	95% C.I.	]
fam_size	.485418	.0588269	8.25	0.000	2.50255	.37012	.600717	
_Isex_ref_2*	.3822398	.1497684	2.55	0.011	0 --> 1	.088699	.67578	
age_ref	.0129315	.0046709	2.77	0.006	50.1176	.003777	.022086	
vehq	.0899312	.0736093	1.22	0.222	1.90058	-.05434	.234203	
_cons	-3.066429	.3441995	-8.91	0.000	1	-3.74105	-2.39181	

Marginal Effects: Probability Uncensored

variable	dF/dx	Std. Err.	z	P> z	X_at	[	95% C.I.	]
fam_size	.0556681	.0067463	8.25	0.000	2.50255	.042446	.068891	
_Isex_ref_2*	.0437949	.0171756	2.55	0.011	0 --> 1	.010131	.077458	
age_ref	.001483	.0005357	2.77	0.006	50.1176	.000433	.002533	
vehq	.0103134	.0084416	1.22	0.222	1.90058	-.006232	.026859	
_cons	-.3516605	.0394731	-8.91	0.000	1	-.429026	-.274295	

(*) dF/dx is for discrete change of dummy variable from 0 to 1

1. Is the tobit model appropriate here?
2. How many household did not buy any sweets?
3. Interpret the effect of sex of the reference person on sweets expenditures.
4. Does additional person in the household affect sweets expenditures?

Exercise 2

On the basis of the Consumer Survey conducted by the Bureau of Labor Statistics, US Department of Labour your lecturer estimated a tobit model for fresh fruit expenditures (*frshfrut*). The list of regressors consisted of:

- *fam_size* – family size
- *age_ref* – age of the reference person in the household (1 for men, 2 for women)
- *age_ref* – age of the reference person in the household

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. xi: tobit frshfrut age_ref fam_size i.sex_ref, ll
i.sex_ref      _Isex_ref_1-2      (_Isex_ref_1 for sex_ref==1 omitted)
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Tobit regression	Number of obs	=	3392
	LR chi2(3)	=	180.03
	Prob > chi2	=	0.0000
Log likelihood = -8655.2198	Pseudo R2	=	0.0103

frshfrut	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
age_ref	.1109201	.0178551	6.21	0.000	.0759122	.145928
fam_size	2.720853	.2071222	13.14	0.000	2.314756	3.12695
_Isex_ref_2	.5987691	.583082	1.03	0.305	-.544459	1.741997
_cons	-12.39072	1.276864	-9.70	0.000	-14.89422	-9.887223
/sigma	15.32552	.2748058			14.78672	15.86433

Obs. summary: 1562 left-censored observations at frshfrut<=0
 1830 uncensored observations
 0 right-censored observations

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. dtobit
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Marginal Effects: Latent Variable

variable	dF/dx	Std. Err.	z	P> z	X_at	[	95% C.I.	]
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age_ref	.1109201	.0178551	6.21	0.000	50.755	.075925	.145915	
fam_size	2.720853	.2071222	13.14	0.000	2.48408	2.3149	3.1268	
_Isex__2*	.5987691	.583082	1.03	0.304	0 --> 1	-.544051	1.74159	
_cons	-12.39072	1.276864	-9.70	0.000	1	-14.8933	-9.88812	

Marginal Effects: Unconditional Expected Value

variable	dF/dx	Std. Err.	z	P> z	X_at	[	95% C.I.	]
-----+-----								
age_ref	.0563977	.0090785	6.21	0.000	50.755	.038604	.074191	
fam_size	1.383427	.105312	13.14	0.000	2.48408	1.17702	1.58983	
_Isex__2*	.3040167	.29647	1.03	0.305	0 --> 1	-.277054	.885087	
_cons	-6.300105	.6492256	-9.70	0.000	1	-7.57256	-5.02765	

Marginal Effects: Conditional on being Uncensored

variable	dF/dx	Std. Err.	z	P> z	X_at	[	95% C.I.	]
-----+-----								
age_ref	.0408215	.0065711	6.21	0.000	50.755	.027942	.053701	
fam_size	1.001344	.0762263	13.14	0.000	2.48408	.851943	1.15075	
_Isex__2*	.2201298	.2145893	1.03	0.305	0 --> 1	-.200457	.640717	
_cons	-4.560108	.4699189	-9.70	0.000	1	-5.48113	-3.63908	

Marginal Effects: Probability Uncensored

variable	dF/dx	Std. Err.	z	P> z	X_at	[	95% C.I.	]
-----+-----								
age_ref	.0028867	.0004647	6.21	0.000	50.755	.001976	.003798	
fam_size	.0708112	.0053904	13.14	0.000	2.48408	.060246	.081376	
_Isex__2*	.0155828	.0151749	1.03	0.304	0 --> 1	-.01416	.045325	
_cons	-.3224734	.0332309	-9.70	0.000	1	-.387605	-.257342	

(*) dF/dx is for discrete change of dummy variable from 0 to 1